MAE 3270 HW 10 Q5 Dowling Problem 7.22 Part (b)

Using von Mises Theory (max octahedral shear stress theory) Prof Alan Zehnder Michelle Zhou Nov 3, 2021

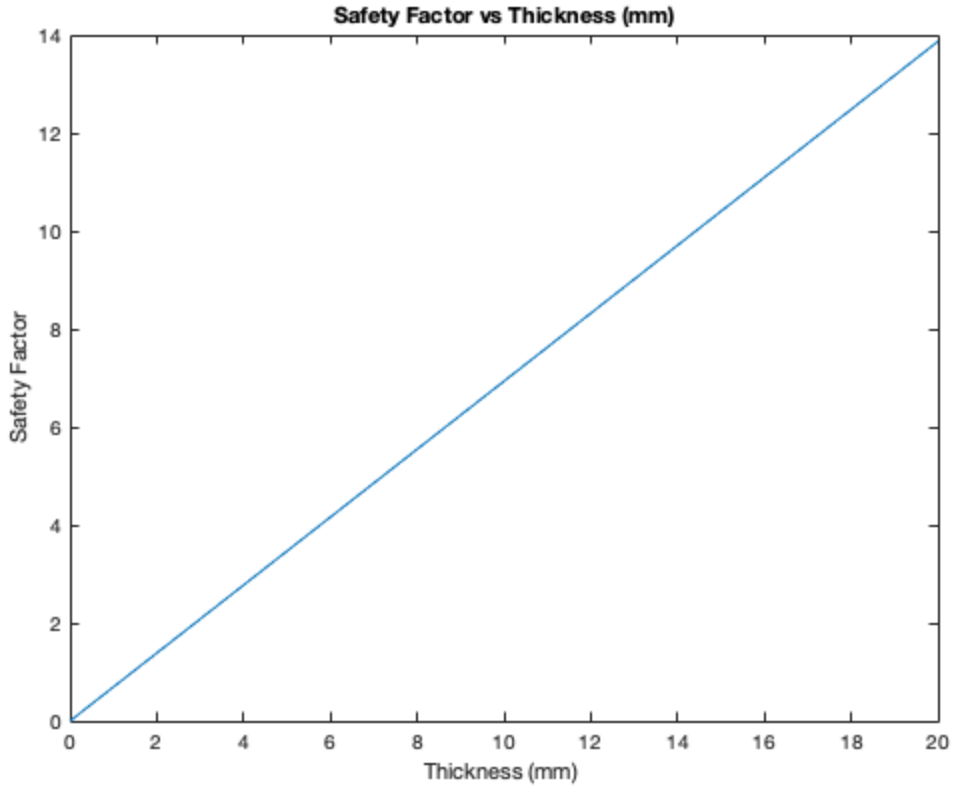
```
P = 50*10^3;            % load in N
r_avg = 0.02525;        % average radius in m
T = 700;                % torque in Nm
sigma_oT = 303*10^6;    % yield stress in tension in Pa

t = [0:0.0001:0.02];    % thickness list in m
A = 2*pi*r_avg*t;       % area list in m^2
J = 2*pi*r_avg^3*t;     % polar moment of inertia list in m^4

sigma_x = P./A;         % normal stress list in the x-dir in Pa
tau_xy = T*r_avg./J;    % shear stress list xy in Pa
sigma_VM = (1/sqrt(2))*sqrt(2*(sigma_x.^2) + 6*(tau_xy.^2)); % von
 Mises stress in Pa

X = sigma_oT./sigma_VM; % dimensionless factor of safety

% plotting safety factor vs thickness
close all
plot(t*10^3, X)
xlabel('Thickness (mm) ')
ylabel('Safety Factor')
title('Safety Factor vs Thickness (mm)')
```



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